Asymptotic behaviour of the Bardeen–Cooper–Schrieffer theory of superconductivity

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Bardeen–Cooper–Schrieffer (BCS) 1957 theory of superconductivity

Described by (multi-band) **BCS gap equation**

$$\Delta_a(p) = -\frac{1}{(2\pi)^{d/2}} \sum_{b=1}^{n} \int_{\mathbb{R}^d} \hat{V}_{ab}(p-q) \frac{\Delta_b(q)}{K_{T,b}^{\Delta_b}} \, \mathrm{d}q \,, \quad a = 1, \ldots, n, \quad K_{T,a}^{\Delta_a}(p) = \frac{\sqrt{\varepsilon_a(p)^2 + |\Delta_a(p)|^2}}{\tanh\left(\frac{\sqrt{\varepsilon_a(p)^2+|\Delta_a(p)|^2}}{2T}\right)}.$$

n: Number of bands (for most of the talk $n = 1$)

V_{ab}: Effective interactions. **Possibly attractive**, meaning V has negative part,

ε_a: Effective dispersion relations, e.g. Sommerfeld: $\varepsilon_a(p) = p^2/2m_a - \mu_a$.

Material is superconducting if non-zero solution $\Delta = (\Delta_a)_{a=1}^n$ exists. In matrix-valued form:

$$\Delta = -\frac{1}{(2\pi)^{d/2}} \hat{V} * (K_T^{\Delta})^{-1} \Delta, \qquad K_T^{\Delta} = \mathrm{diag}[K_{T,a}^{\Delta_a}].$$

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Arises as Euler–Lagrange equation of **BCS functional** (with $\Delta_a = -2 \sum_b \widehat{V_{ab}\alpha_b}$)

$$\mathcal{F}(\gamma, \alpha) = \int_{\mathbb{R}^d} \sum_{a=1}^{n} \varepsilon_a(p) \gamma_a(p) \, \mathrm{d}p + \int_{\mathbb{R}^d} \sum_{a,b=1}^{n} V_{ab}(x) \overline{\alpha_a(x)} \alpha_b(x) \, \mathrm{d}x - TS(\gamma, \alpha).$$

Theorem (Hainzl–Hamza–Seiringer–Solovej 2008)

Let $V_{ab} = V_{ba} \in L^{p_d}$ *be real-valued and even* ($p_3 = 3/2$, $p_2 > 1$, $p_1 = 1$), $\mu \in \mathbb{R}$ *and* $0 \leq T < \infty$. *Then the following are equivalent*

1. *A minimizer of the BCS functional has* $\alpha \neq 0$,
2. *There exists* $\alpha \neq 0$ *with* $(K_T^\Delta + V)\alpha = 0$, *(equivalently* $\Delta = -2\widehat{V\alpha}$ *satisfies the BCS gap equation),*
3. *The (matrix-valued) operator* $K_T^0 + V$ *has at least one negative eigenvalue.*

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$\rightsquigarrow$ Define the **critical temperature** $T_c := \inf\{T \geq 0 : K_T^0 + V \geq 0\}$.

Phase transition at T_c: superconducting for $T < T_c$ and normal metal for $T \geq T_c$. (Follows easily by K_T monotone increasing in T.)

As operators $K_T^0 \geq 2T$. $\rightsquigarrow$ Need V non-positive (as matrix-valued operator) for $T_c > 0$.

$E_\Delta(p) = \sqrt{(p^2 - \mu)^2 + |\Delta(p)|^2}$ dispersion relation for associated quasi-particles.

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T_c and Ξ are material-dependent (μ, V-dependent). Both measure "stability" of superconducting phase.

Question

- Formula for T_c and Ξ in various limits?
- Relation between T_c and Ξ ?

Zero temperature

Physics prediction for T_c small:

$$\frac{\Xi(T=0)}{T_c} \approx \pi e^{-\gamma} \approx 1.76$$

Well-known in physics literature.

Valid for s-wave superconductivity (Δ radial),
(Ex: V radial with $\hat{V} \leq 0$)

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Simple regimes with T_c small:

- Weak coupling λV with $\lambda \to 0$,
- Low density $\mu \to 0$,
- High density $\mu \to \infty$.

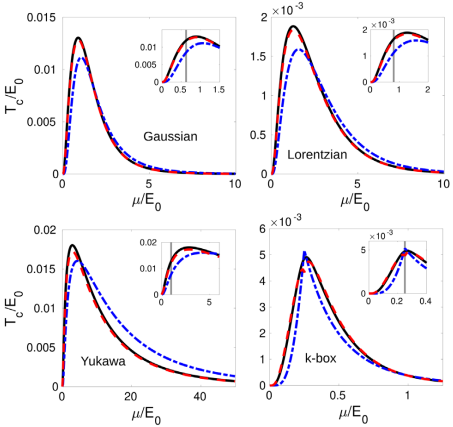

Figure: From "E. Langmann, C. Triola, and A. V. Balatsky. Phys. Rev. Lett. **122**.15 (2019)"

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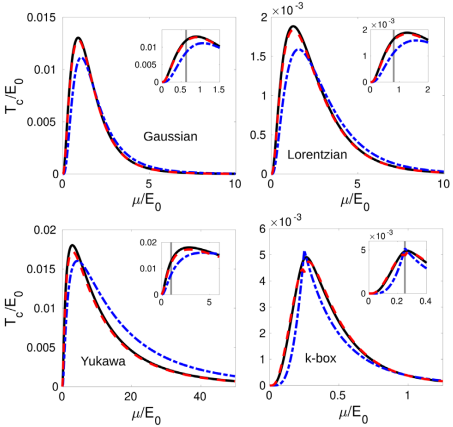
Question

Can we find T_c and $\Xi(T=0)$ in these limits?



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Zero temperature

Theorem

Suppose V radial and $\hat{V} \leq 0$ (s-wave superconductivity). Then, T_c and $\Xi(T=0)$ satisfy

$$\frac{\Xi}{T_c} = \pi e^{-\gamma}(1 + o(1))$$

- *in dimensions $d = 1, 2, 3$ for interaction λV in limit $\lambda \to 0$ (weak coupling)* [Frank–Hainzl–Naboko–Seiringer 2007; Hainzl–Seiringer 2008b, Henheik–L.–Roos 2024],
- *in dimension $d = 3$ in limit $\mu \to 0$ (low density)* [Hainzl–Seiringer 2008a, L. 2021],
- *in dimension $d = 3$ in limit $\mu \to \infty$ (high density)* [Henheik 2022, Henheik–L. 2022],
- *in dimension $d = 3$ in "any" limit $T_c, \Xi \to 0$* [Langmann–Triola–Balatsky 2019; Langmann–Triola 2023].

Proofs by separate asymptotic formulas for T_c and Ξ in the various limits.

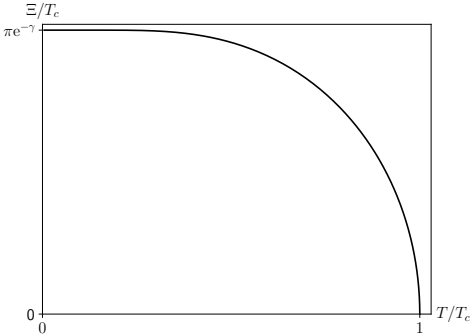
Non-zero temperature

Physics prediction for T_c small (for s-wave superconductivity):

$$\frac{\Xi(T)}{T_c} \approx \mathrm{f}_{\mathrm{BCS}}\left(\sqrt{1-\frac{T}{T_c}}\right)$$

for **universal** function $\mathrm{f}_{\mathrm{BCS}}$. For $T \to T_c$:

$$\frac{\Xi(T)}{T_c} \approx \sqrt{\frac{8\pi^2}{7\zeta(3)}}\sqrt{1-\frac{T}{T_c}} \approx 3.06\sqrt{1-\frac{T}{T_c}}.$$



Theorem (Langmann–Triola 2023, Henheik–L. 2025)

Picture holds

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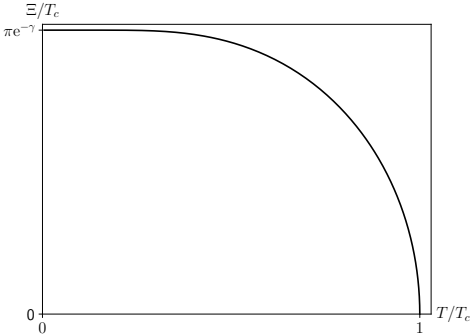

Theorem (Langmann–Triola 2023, Henheik–L. 2025)

Picture holds for interaction λV radial in limit $\lambda \to 0$

- *in dimensions $d = 1, 2, 3$ for V with $\hat{V} \leq 0$* (s-wave superconductivity, weak coupling limit),
- *in dimension $d = 2$ for V with* $\dim\ker(K_{T_c} + \lambda V) \leq 2$ (pure angular momentum, weak coupling limit),

Multi-band effects

To probe **inter-band** effects we introduce coupling parameters ($V^{\mathrm{d}}_{ab} = \delta_{ab} V_{aa}$, $V^{\mathrm{od}}_{ab} = (1-\delta_{ab}) V_{ab}$,)

$$V \longrightarrow \lambda V^{\mathrm{d}} + \kappa \lambda V^{\mathrm{od}}, \qquad \lambda > 0, \quad \kappa \in \mathbb{R}.$$

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Problem

Study limit of **weak coupling and weaker inter-band coupling**: $T_c(\lambda, \kappa)$ *for* $\lambda, |\kappa| \ll 1$.

Important role played by operators "V on the Fermi surface(s)"

$$\mathcal{V}_{ab} = \mathfrak{F}_a V_{ab} \mathfrak{F}_b^{\dagger}, \qquad \mathfrak{F}_a\text{: Fourier transform restricted to } S_a = \{p : \varepsilon_a(p) = 0\}.$$

$$\text{Group in} \quad \mathcal{V} = \mathcal{V}^{\mathrm{d}} + \kappa \mathcal{V}^{\mathrm{od}} \quad \text{and define} \quad \mathfrak{e}_a := \inf \operatorname{spec} \mathcal{V}_{aa}.$$

Theorem (Frank–Hainzl–Naboko–Seiringer 2007)

Let $\kappa = 0$ *and suppose* $\mathfrak{e}_{\min} = \min_a \mathfrak{e}_a < 0$. *Then,* $T_c(\lambda, 0) > 0$ *for all* $\lambda > 0$ *and*

$$T_c(\lambda, 0) \sim \exp\left(\frac{1}{\lambda \mathfrak{e}_{\min}}\right) \qquad \textit{for} \qquad \lambda \to 0.$$

Theorem (Henheik–Langmann–L. 2024)

Suppose $\min_{a\in\{1,\dots,n\}} \mathfrak{e}_a < 0$. *Then, for* $\lambda, |\kappa| \ll 1$*:*

(linear) *If at least 2 minimizers* $\hat{a}_1, \hat{a}_2 \in \{1, \dots, n\}$ *of* $a \mapsto \mathfrak{e}_a$*:*

$$T_c(\lambda,\kappa) = T_c(\lambda,0)\exp\left[\frac{A_1^{\operatorname{sgn}(\kappa)}}{\lambda^2}|\kappa\lambda|\left[1+O(\lambda)+O(\kappa)\right]\right], \quad A_1^{\pm} > 0.$$

If exactly 2 minimizers $\hat{a}_1, \hat{a}_2$ *and* $\dim\ker(\mathcal{V}_{\hat{a}_i\hat{a}_i} - \mathfrak{e}_{\hat{a}_i}) = 1$ *for at least one* $i = 1, 2$*, then* $A_1^+ = A_1^-$*. (Otherwise generically* $A_1^+ \neq A_1^-$*.)*

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(quadratic) *If minimizer* $\hat{a}$ *of* $a \mapsto \mathfrak{e}_a$ *is unique:*

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Recall $\quad T_c \sim \exp\left(1/\lambda\mathfrak{e}_{\min}(\kappa)\right), \quad \mathfrak{e}_{\min}(\kappa) = \inf\operatorname{spec}\mathcal{V}(\kappa), \quad \mathcal{V}(\kappa) = \mathcal{V}^{\mathrm{d}} + \kappa\mathcal{V}^{\mathrm{od}}.$

Example (Idea behind 'linear' versus 'quadratic')

Consider eigenvalues of 2×2 matrices

$$\begin{bmatrix}1 & \kappa\\ \kappa & 1\end{bmatrix}, \quad \lambda_{\max} = 1+\kappa, \qquad \text{and} \qquad \begin{bmatrix}1 & \kappa\\ \kappa & 0\end{bmatrix}, \qquad \lambda_{\max} = \frac{1+\sqrt{1+4\kappa^2}}{2} = 1+\kappa^2+O(\kappa^4).$$

Main ideas of proofs — Birman–Schwinger principle

"$\Delta \neq 0$ solves BCS gap equation"

$\Leftrightarrow$ "$K_T^\Delta + V$ has lowest eigenvalue 0 and associated eigenvector $\alpha = -\frac{\Delta}{2K_T^\Delta}$"

$\Leftrightarrow$ "Birman–Schwinger operator

$$B_{T,\Delta} = \operatorname{sgn} V |V|^{1/2} (K_T^\Delta)^{-1} |V|^{1/2}$$

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To study $B_{T,\Delta}$ define

$$m(T,\Delta) = \frac{1}{|\mathbb{S}^{d-1}|} \int_{|p| \leq \sqrt{2\mu}} \frac{1}{K_T^\Delta(p)} \, \mathrm{d}p$$

Decompose (with $\mathfrak{F}_\mu : L^1(\mathbb{R}^d) \to L^2(\mathbb{S}^{d-1})$ a rescaled Fourier transform (here $\varepsilon_a(p) = p^2/2m_a - \mu_a$))

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$B_{T,\Delta}$ has -1 as lowest eigenvalue $\rightsquigarrow$

$$m(T,\Delta) = -\left[\inf \operatorname{spec} \left(\mathfrak{F}_\mu |V|^{1/2} \frac{1}{1 + \operatorname{sgn} V |V|^{1/2} M_{T,\Delta} |V|^{1/2}} \operatorname{sgn} V |V|^{1/2} \mathfrak{F}_\mu^\dagger \right) \right]^{-1}.$$

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Evaluate directly $m(T,\Delta) = \frac{1}{|\mathbb{S}^{d-1}|}\int_{|p|\leq\sqrt{2\mu}} \frac{1}{K_T^\Delta(p)}\,\mathrm{d}p$ at $T=0$ and $T=T_c$:

$$m(T=0,\Delta) \approx \mu^{d/2-1}\left[\log\frac{\mu}{\Xi(T=0)} + \log(2c_d)\right], \quad m(T_c,\Delta=0) \approx \mu^{d/2-1}\left[\log\frac{\mu}{T_c} + \log\frac{e^\gamma}{\pi} + \log(2c_d)\right],$$

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Zero-temperature: Neglect error term $M_{T,\Delta}$ and consider for $T=0$ and $T=T_c$:

$$m(T=0,\Delta) \approx m(T_c,\Delta=0) \approx \frac{-1}{\mathfrak{e}_\mu}, \qquad \mathfrak{e}_\mu = \inf\operatorname{spec}\left(\mathfrak{F}_\mu V \mathfrak{F}_\mu^\dagger\right) < 0.$$

Thus,

$$T_c \approx 2c_d\frac{e^\gamma}{\pi}\exp\left(\frac{1}{\mu^{d/2-1}\mathfrak{e}_\mu}\right), \qquad \Xi(T=0) \approx 2c_d\exp\left(\frac{1}{\mu^{d/2-1}\mathfrak{e}_\mu}\right),$$

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Positive temperature: Neglect error term $M_{T,\Delta}$ and consider for T and $T=T_c$:

$$\int_0^{\sqrt{2\mu}}\left(\frac{1}{K_T^\Delta(p)} - \frac{1}{K_{T_c}^0(p)}\right)|p|^{d-1}\,\mathrm{d}p = m(T,\Delta) - m(T_c,\Delta=0) \approx 0$$

Essentially defining equation for $\mathrm{f}_{\mathrm{BCS}} \rightsquigarrow \Xi(T)/T_c \approx \mathrm{f}_{\mathrm{BCS}}(\sqrt{1-T/T_c})$.

$$m(T,\Delta) = -\left[\inf\operatorname{spec}\left(\mathfrak{F}_\mu |V|^{1/2} \frac{1}{1+\operatorname{sgn} V |V|^{1/2} M_{T,\Delta} |V|^{1/2}} \operatorname{sgn} V |V|^{1/2} \mathfrak{F}_\mu^\dagger\right)\right]^{-1},$$

$$m(T_c, \Delta = 0) \approx \mu^{d/2-1}\left[\log\frac{\mu}{T_c} + \log\frac{e^\gamma}{\pi} + \log(2c_d)\right].$$

Multi-band: Neglect error term $M_{T,\Delta}$ and consider for $\kappa = 0$ and $\kappa \neq 0$:

$$\begin{aligned}\mu^{d/2-1}\log\frac{T_c(\kappa)}{T_c(0)} &\approx m(T_c(\kappa=0),\Delta=0) - m(T_c(\kappa),\Delta=0)\\ &\approx -\frac{1}{\inf\operatorname{spec}(\mathfrak{F}_\mu V^{\mathrm{d}} \mathfrak{F}_\mu^\dagger)} + \frac{1}{\inf\operatorname{spec}(\mathfrak{F}_\mu (V^{\mathrm{d}} + \kappa V^{\mathrm{od}}) \mathfrak{F}_\mu^\dagger)}.\end{aligned}$$

$$m(T,\Delta) = -\left[\inf\operatorname{spec}\left(\mathfrak{F}_\mu |V|^{1/2} \frac{1}{1+\operatorname{sgn} V |V|^{1/2} M_{T,\Delta} |V|^{1/2}} \operatorname{sgn} V |V|^{1/2} \mathfrak{F}_\mu^\dagger\right)\right]^{-1},$$

$$m(T_c, \Delta = 0) \approx \mu^{d/2-1}\left[\log\frac{\mu}{T_c} + \log\frac{e^\gamma}{\pi} + \log(2c_d)\right].$$

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Standard perturbation theory in κ.

$\rightsquigarrow$ conditions for "linear" versus "quadratic".